

Module 03: Perception in Robotics — Sensor Processing for Control

Learning Objectives

1. Define robotic **perception for control** as the extraction of task-relevant state information from raw sensor data.
2. Analyze how different sensor modalities (vision, LiDAR, tactile, force/torque) implement the observation model (A matrix) for control applications.
3. Apply the concept of **observation design** — choosing and configuring sensors to maximize the informativeness of observations for the control task.

Introduction

Control requires perception — a robot cannot control what it cannot observe. But perception for control is different from perception in general: it is **task-directed**. A robot arm performing assembly doesn't need to understand the entire visual scene — it needs to know the position and orientation of the part it's assembling. A mobile robot doesn't need to identify every object — it needs to know where obstacles are and where traversable space is.

Key Concepts

1. Sensor Modalities and Their A Matrices

Each sensor modality implements a different observation model:

- **Encoders** (joint position sensors): Simple, direct A matrix — the observation is a noisy linear function of the joint angle. High precision, low ambiguity.
- **IMU** (Inertial Measurement Unit): Provides acceleration and angular velocity. The A matrix involves integrating these to get position and orientation — introducing drift (accumulating noise). Moderate precision, needs fusion.

- **Camera:** Rich, high-dimensional observations (millions of pixels). The A matrix involves 3D-to-2D projection, lighting variation, and occlusion. High ambiguity — many different 3D scenes can produce similar 2D images.
- **LiDAR:** Provides direct distance measurements in a scan pattern. The A matrix is geometric (ray casting). Low ambiguity for geometry, but no color or texture information.
- **Force/Torque sensors:** Measure contact forces at end-effectors. The A matrix relates hidden contact states (friction, material stiffness, contact geometry) to observed force vectors.

2. Feature Extraction as Dimensionality Reduction

Raw sensor data is high-dimensional; control requires low-dimensional state estimates. Feature extraction bridges this gap:

- **Visual features:** Edges, corners, SIFT/ORB keypoints, deep-learned features — reduce a megapixel image to hundreds of task-relevant descriptors
- **Point cloud processing:** Down-sampling, plane fitting, object segmentation of LiDAR data — reduce millions of 3D points to geometric primitives
- **Tactile feature extraction:** Contact location, normal force, friction coefficient from distributed tactile arrays

In Active Inference terms, feature extraction is **model-based dimensionality reduction** — the generative model specifies which aspects of the observation are relevant for the current inference task, and the feature extractor computes sufficient statistics.

3. Observability and Sensor Selection

Not all states are observable from all sensors. **Observability analysis** determines which states can be estimated from the available observations:

- A camera alone cannot observe depth at a single frame (unobservable) — but with structure from motion (multiple frames + known motion), depth becomes observable
- An encoder alone cannot detect external contact forces — but combined with a dynamics model and motor current measurements, contact can be inferred

- **Sensor selection** is the engineering decision about which A matrices to implement — choosing sensors that make the task-relevant states observable

4. Active Perception for Control

When perception is insufficient for the current control task, the robot can take **epistemic actions** — actions chosen to improve state estimation:

- A robot arm unsure about an object's pose can rotate the object to bring distinguishing features into view
- A mobile robot unsure about its location can move to a landmark-rich area to improve localization
- A manipulation robot unsure about an object's weight can lift it slightly to generate diagnostic force observations

These are Active Inference epistemic actions — reducing expected ambiguity through deliberately informative behavior.

Applications

- **Visual servoing:** A robot arm uses real-time camera feedback to align with a target — the controller directly minimizes visual prediction error (the difference between the observed and desired image features), bypassing explicit 3D state estimation entirely.
- **Tactile slip detection:** A gripper monitors shear force patterns to detect incipient slip — when the tactile prediction error exceeds a threshold (force pattern deviates from stable-grasp model), the controller increases grip force before the object falls.

Conclusion

Perception for control transforms raw sensor data into task-relevant state estimates through sensor-specific A matrices, feature extraction, observability analysis, and active perception. The next module examines the cognitive processing that builds on these perceptual estimates.